

each row and column pin

A delay function (delay(t)) is introduced to provide a time gap between steps, helping to control the display timing.



Fig. 6: Pick and place machine



Fig. 7: Reflow oven

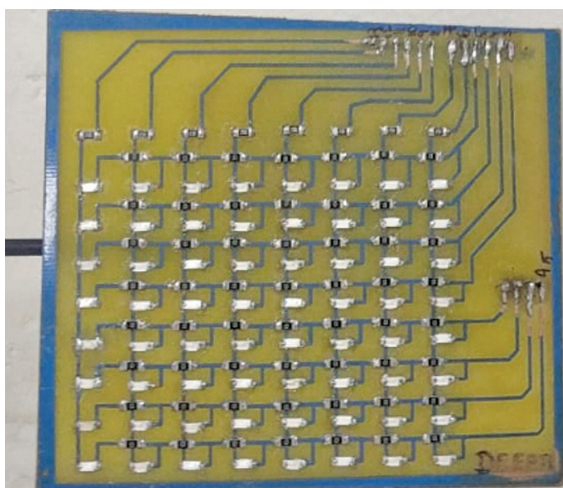


Fig. 8: Authors' PCB prototype

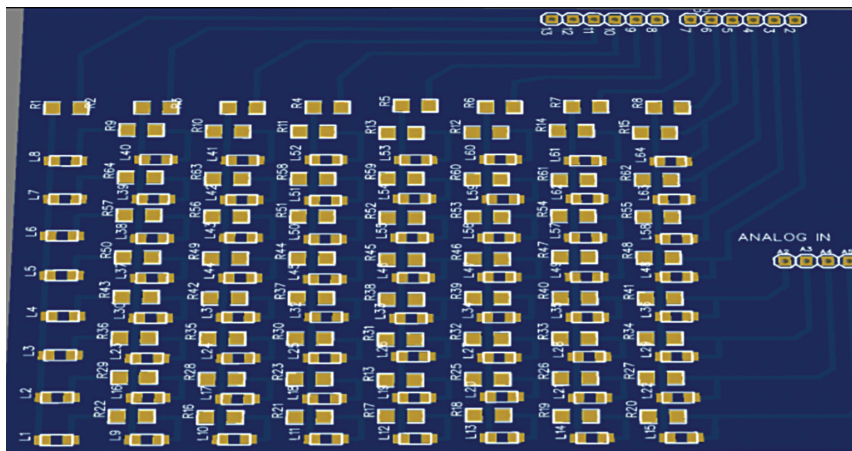


Fig. 9: Actual-size 3D model of the board

Construction and testing

The 8×8 matrix display interfaced with the Arduino Uno is constructed using 64 SMD 1206 LEDs and $1k\Omega$ resistors. To avoid the intersection of circuit lines, 0Ω resistors are used. Fig. 3 shows the circuit diagram of the 8×8 SMD matrix display interfaced with the Arduino Uno.

The first stage in developing the SMD PCB for the 8×8 matrix display involves designing the PCB layout using EDA software. The layout is then exported as a PNG, as shown in Fig. 4, and imported into EzCAD for laser etching on an FR4 board. Using a laser etching machine, the bare PCB for the matrix display is fabricated, as shown in Fig. 5.

The second stage involves assembling the components. After cleaning the etched board, a generous amount of solder paste is applied to the solder pads. Using a pick-and-place machine (see Fig. 6), the components are then mounted in their corresponding positions.

The final stage of PCB fabrication is soldering. In this stage, a reflow oven is used to solder the components using the hot-air reflow

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method. The five chambers of the reflow oven (see Fig. 7) are set with progressively increasing temperatures and controlled airflow to ensure a proper soldering finish.

Male-to-male connectors are hand-soldered for interfacing with the Arduino Uno. After completion of the soldering process, the circuit should be carefully inspected to ensure there are no short or open connections. Once verified, the PCB is connected to the Arduino and the program uploaded to visualise the output. Fig. 8 shows the author's PCB prototype, while Fig. 9 illustrates the actual-size 3D model of the board. **EFY**

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